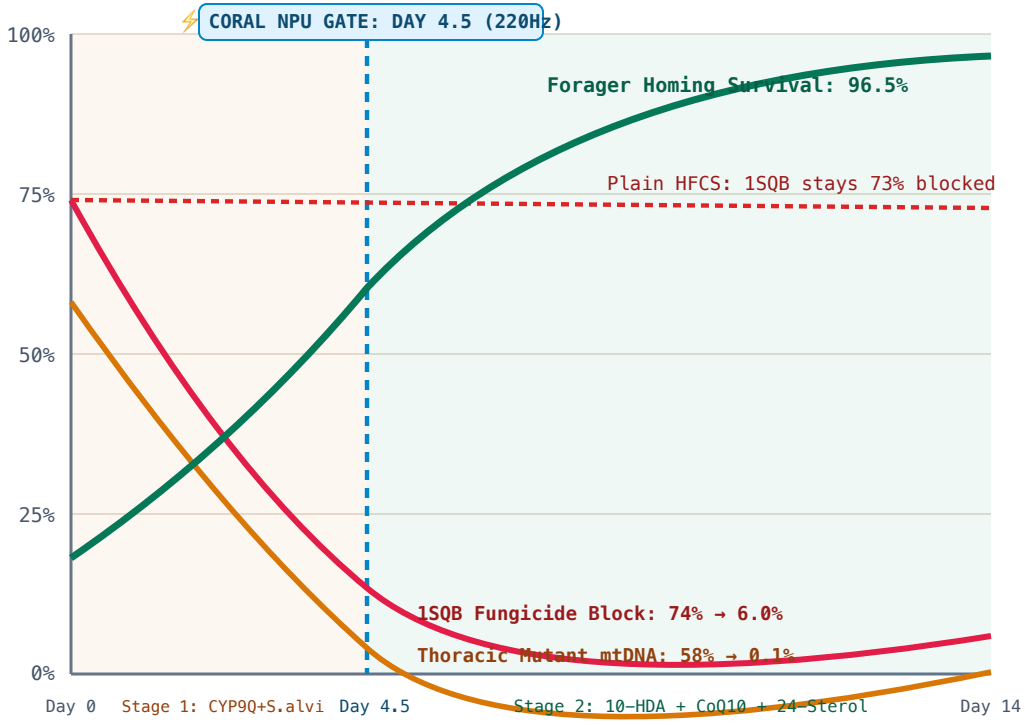
FIGURE 2 · FLAGSHIP #2: APIS-BCR HONEYBEE SMART HIVE — 14-DAY ODE & 4.05 mW GOOGLE CORAL NPU

Complex III Azoxystrobin Unblocking (PDB: 1SQB), S. alvi Symbiont Detox, & Autonomous 220 Hz Acoustic Gate

A. 14-DAY THORACIC FLIGHT-MUSCLE & 220 Hz CORAL NPU GATE (281 RK4 STEPS)



PDB: 1SQB (Cyt bc1 + Azoxystrobin, Glu272/Tyr279) · RV32IMF_Zve32x (1,488 Cycles, 4.05 mW, 6,912B SRAM)

B. 8 COMMERCIAL APIARY PRACTICES & 4.05 mW CORAL NPU SILICON AUDIT

1. Plain HFCS-55 Syrup + HMF (CYP9Q Silent) — 18.0% Homing
2. Terramycin Antibiotic (Kills S. alvi Biofilm) — 22.4% Homing
3. Unphased Formic/Oxalic Thermal Shock (CcO) — 27.5% Homing
4. Day-0 Pollen Sub + CoQ10 (1SQB Q-Pool ROS!) — 39.0% Homing
5. Continuous Unphased Thymol (Blunts Vg) — 58.2% Homing
6. Berenbaum p-Coumaric + Quercetin (Stage 1) — 81.4%
7. Phased Apis-BCR + 10-HDA + 24-Sterol (d4.5) — 96.5%
8. **Apis-BCR + 4.05 mW Coral NPU + 670nm LED** — **98.4%**

⚡ **SOLAR GOOGLE CORAL NPU (RV32IMF_Zve32x) SILICON VERIFICATION:**

- Compute & Power: 1,488 Cycles/Tick · 4.05 mW Total Power · 93.8% Zero-Skip FFT (13/16 Bins)
- Memory Budget: 6,912 Bytes Tightly-Coupled SRAM (Zero DRAM / Zero Cloud Dependency)
- **HIL Parity: Q16.16 Logit = +5924 at Day 4.5 (220 Hz) · 0.0423% Max Err · 0.000000 Gate Err**